Task 4: Quality, Deployment and Compliance

Final Project - Task 4 | Digital Loan Pre-Approval & Onboarding (MVP), Online Bank | Prepared by: [Your Name], Business Analyst | 09 November 2026

Part 1: Test Cases

> **AI prompt used:** *Generate 8 test cases for a digital loan onboarding system at an online bank using Gherkin format, based on these user stories and business rules: [US-01 to US-10, BR-01 to BR-10]. Include positive tests for successful onboarding and negative/edge tests for KYC failure, missing data, borderline credit score, and credit API timeout. Give each a Test ID, description, preconditions, Gherkin steps and expected result.*

Test ID	Description	Preconditions	Steps (Gherkin)	Expected result
TC-01	Successful submission and pre-approval (positive)	Customer registered; consent given; test identity passes KYC; bureau stub returns score 720, DTI 30%	Given a customer with valid identity and consent When they submit a complete application for $10,000 over 36 months Then KYC returns "Verified", a soft pull is recorded And the decision "Pre-approved" is shown within 3 minutes	Status = Pre-approved; offer displayed; notification sent; audit entries for consent, KYC, pull, decision
TC-02	Missing mandatory data (negative)	Consent given	Given the date of birth field is empty When the customer submits the application Then status = "Incomplete" and the field is highlighted And no KYC or credit call is made	Incomplete; no third-party calls (BR-01)
TC-03	KYC failure (negative)	Test identity configured to fail	Given a submitted application When the KYC Service returns "no match" Then status = "KYC failed" and no credit pull occurs	KYC failed; customer told how to contact support; logged
TC-04	KYC partial match (edge)	Identity with address mismatch	Given a partial KYC match When the result is processed Then the case is routed to the manual review queue with reason "KYC partial match"	Case in queue; SLA timer started (BR-03)
TC-05	Borderline credit score routing (edge)	KYC passed; bureau stub score 650	Given a verified application When the soft pull returns 650 Then the decision engine routes the case to manual review	Manual review; no auto decision (BR-05)
TC-06	Low score decline (negative)	KYC passed; score 560	Given a verified application When the soft pull returns 560 Then status = "Declined" with reason codes And an adverse-action notice is sent	Declined; notice sent (BR-09)
TC-07	Credit API timeout with retry (error handling)	Bureau stub set to time out	Given a verified application When the Credit Bureau API times out 4 times Then the system retries 3 times with backoff And sets status "Pending - service delay" and notifies the customer	No decline; routed to review; 3 retries logged (BR-06)
TC-08	Credit pull blocked without consent (negative / compliance)	Soft-pull consent not recorded	Given an application without soft-pull consent When the system attempts a credit pull Then the pull is blocked and consent is requested again	No bureau call; event logged (BR-02)
TC-09	Audit log immutability (compliance)	Admin test account	Given an existing audit entry When any user attempts to edit or delete it Then the action is rejected and the attempt is itself logged	Entry unchanged; tamper attempt recorded

Coverage: positive (TC-01), negative (TC-02, 03, 06, 08), edge (TC-04, 05), error handling (TC-07), compliance (TC-08, 09). Each test traces to the user stories in Task 1 and the business rules in Task 3.

Part 2: User Acceptance Testing (UAT) Plan

Scope

- End-to-end onboarding on web and mobile: start, consent, KYC, soft pull, decision, notification
- Business rule accuracy (BR-01 to BR-10) using prepared test personas
- Integration reliability with the KYC sandbox, bureau sandbox, Core Banking test and notification services
- User experience: completion under 10 minutes, clear messages and accessibility
- Manual review console for Operations

Out of scope: load/performance testing (done in the performance test phase) and features outside the MVP.

Roles and responsibilities

Role	Responsibility
Product Owner	Approves business functionality; final UAT sign-off
QA Lead	Plans and manages test execution, defects and reporting
Compliance Reviewer	Validates consent capture, audit logs, notices and data handling
Operations Analyst	Verifies the manual review queue, SLA timers and agent workflow
Business Analyst	Prepares scenarios, traces results to requirements, clarifies rules
Dev / IT Lead	Fixes defects, maintains the UAT environment

Schedule

Two weeks: week 1 functional and rule scenarios; week 2 end-to-end, regression of fixes, and sign-off.

Entry and exit criteria

Entry criteria	Exit criteria
Requirements and business rules signed off	95% of test cases passed
UAT environment ready with vendor sandboxes	No open Critical or High defects
Test data prepared (synthetic personas, no real customer data)	Compliance checks complete and signed
System test passed; release notes available	Product Owner and Compliance sign-off recorded

Defect tracking and SLA

Severity	Definition	Resolution SLA	Re-test
Critical	Blocks onboarding, wrong decision, or compliance/data breach	Fix within 24 hours	Immediate re-test + regression of affected flow
High	Major function fails with a workaround	Fix within 2 business days	Re-test within 1 day of fix
Medium	Partial function or confusing message	Fix within 5 business days	Re-test in next cycle
Low	Cosmetic or minor text issue	Next release	Optional

Defects are logged in Jira with steps, evidence, severity and linked test case/requirement, and reviewed in a daily triage meeting.

Part 3: Deployment Readiness Checklist

Pre-deployment

- ☐ Code reviewed, approved and merged; release tagged
- ☐ All automated tests and UAT test cases passed; sign-offs attached

- ☐ Feature flags configured (new onboarding OFF by default; canary group defined)
- ☐ Production credentials and API keys stored in a secrets vault; access reviewed
- ☐ Database migrations tested on a production-like copy; backup taken
- ☐ Monitoring dashboards and alerts enabled (latency, error rate, decision mix, API failures)
- ☐ Audit logging verified in production configuration
- ☐ Rollback plan rehearsed; on-call roster and communication plan confirmed
- ☐ Customer Support briefed with FAQs and scripts

Post-deployment smoke tests

- ☐ Confirm API Gateway health and authentication
- ☐ Validate KYC and Credit Bureau API connectivity (test identity, soft pull)
- ☐ Test customer login on web and mobile
- ☐ Submit a test loan application end-to-end and confirm decision, notification and audit entries
- ☐ Confirm the Core Banking record is created and the manual review console loads

Rollback strategy

Item	Plan
When to roll back	Critical defect affecting decisions or data; error rate > 5% for 10 minutes; KYC/credit integration failing; any suspected data exposure
Who authorizes	Release Manager with Product Owner (Compliance consulted for data issues); on-call engineer may disable the feature flag immediately
Automated rollback	Turn off the feature flag (instant), then automated redeploy of the previous version via the CI/CD pipeline
Manual rollback	Revert database migration scripts; re-route traffic; validate with smoke tests
Data recovery	Point-in-time restore (RPO 15 min); in-flight applications marked "Pending" and re-processed; customers notified
Communication	Status update to stakeholders within 30 minutes; incident review within 5 business days

Part 4: Compliance, Performance and Scalability Brief

Assumes a US regulatory context; final applicability to be confirmed by Compliance and Legal.

Compliance

- **Consent and credit reporting (FCRA):** soft pull only with recorded consent and a permissible purpose; adverse-action notices with reasons and bureau details
- **Fair lending (ECOA / Reg B):** rules use only credit-related factors; rule and reason codes reviewed for disparate impact
- **KYC / CIP and AML (BSA):** identity verification and sanctions screening before approval
- **Privacy (GLBA) and e-sign (E-SIGN Act):** privacy notice, versioned disclosures, data minimization, encryption and access control
- **Audit:** immutable logs of consent, data access, decisions and rule versions; retention 7 years (to confirm)

Performance

- Automated decision under 3 minutes for 95% of applications; API p95 < 2 s
- Third-party calls with 10 s timeout, 3 retries and circuit breaker
- Performance test at 3x expected peak (50 applications/minute) before launch

Scalability and resilience

- Stateless services behind the API Gateway with auto-scaling
- Queue-based processing for KYC and credit calls so spikes do not drop applications

- 99.9% availability; RTO 4 hours, RPO 15 minutes; multi-zone deployment
- Monitoring and alerting on latency, error rate, review-queue size and decision mix